LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Daily Practice: Ch 5 Life Processes

Chapter / Topic: Life Processes · Week 3 of the 90-day plan

For teacher use only. Do not distribute to students.

DAY 1: NUTRITION

Q1. Which of these organisms is a parasite?

Answer: (c) Cuscuta: it takes its food from the host plant it grows on

Q2. The enzyme in saliva (salivary amylase) breaks down:

Answer: (b) starch, into a simpler sugar

Q3. Bile is made in the:

Answer: (a) liver. It is stored in the gall bladder.

Q4. During photosynthesis, the oxygen given out comes from the splitting of:

Answer: (d) water

Q5. Describe, step by step, how Amoeba takes in and digests its food.

Answer: Amoeba pushes out finger-like projections (pseudopodia) around the food particle, which fuse to form a food vacuole. The food is broken down inside the vacuole, and the digested food diffuses into the cytoplasm. Undigested material is moved to the surface and thrown out.

Q6. What is peristalsis? Why is it needed?

Answer: Peristalsis is the rhythmic contraction of the muscles lining the alimentary canal. It pushes the food forward, so that it can be processed properly in each part of the canal.

Q7. Give the function of: (a) the villi of the small intestine, (b) pancreatic juice, (c) the sphincter muscle at the exit of the stomach.

Answer: (a) Villi increase the surface area for absorbing digested food; they have a rich blood supply that carries the food to every cell.
(b) Pancreatic juice contains trypsin, which digests proteins, and lipase, which breaks down emulsified fats.
(c) The sphincter controls the release of food from the stomach into the small intestine, a little at a time.

Q8. Desert plants such as cacti keep their stomata closed during the day to save water. How do they still carry out photosynthesis?

Answer: They open their stomata at night and take in carbon dioxide, which they store as an intermediate compound. During the day, with the stomata closed, the energy absorbed by chlorophyll is used to make food from this stored compound.

DAY 2: RESPIRATION AND TRANSPORT

Q9. In yeast, the anaerobic breakdown of pyruvate gives:

Answer: (c) ethanol, carbon dioxide and energy (fermentation)

Q10. The respiratory pigment haemoglobin is found in:

Answer: (a) red blood cells: it carries oxygen from the lungs to the tissues

Q11. Which blood vessels have thick, elastic walls because the blood in them is under high pressure?

Answer: (b) arteries: they carry blood away from the heart

Q12. The energy currency of the cell is:

Answer: (b) ATP: the energy released in respiration is stored in ATP and used for every activity of the cell

Q13. What is lymph? Give two of its functions.

Answer: Lymph is the colourless fluid that escapes from the blood capillaries into the spaces between cells. It carries digested and absorbed fat from the intestine, and it drains excess fluid from the spaces between cells back into the blood.

Q14. Why are the walls of capillaries only one cell thick?

Answer: So that materials such as oxygen, food and wastes can easily pass between the blood and the surrounding cells.

Q15. What is blood pressure? Name the instrument used to measure it, and give the normal systolic and diastolic values.

Answer: Blood pressure is the force that blood exerts against the wall of a blood vessel. It is measured with a sphygmomanometer. Normal values are about 120 mm of Hg (systolic) and 80 mm of Hg (diastolic). High blood pressure is called hypertension.

Q16. Trace the path of a molecule of oxygen from the air you breathe in to a muscle cell in your leg.

Answer: Nostrils → pharynx → larynx → trachea → bronchi → bronchioles → alveoli. It diffuses into the blood in the capillaries around the alveoli and binds to haemoglobin in red blood cells. Pulmonary vein → left atrium → left ventricle → aorta → arteries → capillaries in the leg muscle. There it diffuses out of the blood into the muscle cell.

DAY 3: EXCRETION, MIXED PRACTICE AND CHALLENGE

Q17. **Assertion (A):** Anaerobic respiration releases more energy than aerobic respiration. **Reason (R):** In aerobic respiration, glucose is broken down…

Answer: (d) A is false: aerobic respiration releases much more energy. R is true, and complete breakdown is the reason aerobic respiration releases more.

Q18. The basic filtration unit of the kidney is the:

Answer: (d) nephron

Q19. Which of these is NOT a job of the kidneys?

Answer: (c) making bile: that is the liver's job

Q20. To find out whether plants need carbon dioxide for photosynthesis, Meera took two similar de-starched potted plants, P and Q, and placed each under a…

(i) Potassium hydroxide absorbs carbon dioxide, so the air around plant Q had no carbon dioxide.
(ii) The leaf from plant P.
(iii) Starch (food) was made only by the plant that had carbon dioxide. So carbon dioxide is necessary for photosynthesis.

Q21. (a) Give three differences between arteries and veins. (b) What would happen if a person's blood had no platelets? (c) A fish heart has only two…

Answer: (a) Arteries carry blood away from the heart; veins bring it back. Arteries have thick, elastic walls; veins have thinner walls. Veins have valves to keep blood flowing one way; arteries do not.
(b) The blood would not clot, so even a small cut would keep bleeding.
(c) Only once: the heart pumps blood to the gills, and from there it goes to the rest of the body (single circulation).

Q22. A disease damages most of the villi in a person's small intestine. What effect would this have on the person, and why?

Answer: The surface area for absorbing digested food would fall sharply, so much of the digested food would not be absorbed into the blood. The person would become weak and lose weight, even while eating enough.